

CHAPTER 2 : PROPOSED SYSTEM

2.1 Proposed System

The proposed system is a Project Management System designed to digitally manage and monitor the complete academic project lifecycle within the institute. The system enables the Project Coordinator (Admin) to create batches, assign faculty mentors, allocate students, and define customizable project stages with deadlines.

The platform replaces the traditional manual process of tracking project submissions, approvals, and evaluations with a centralized, automated, and structured web-based solution. It provides role-based access control where:

- Admin manages batches, stages, deadlines, and overall monitoring.
- Faculty reviews student submissions, provides remarks, and approves or rejects documents.
- Students upload project documents, track progress, and receive feedback.

The proposed system ensures:

- Centralized academic project monitoring
- Structured stage-wise submission tracking
- Secure role-based authentication
- Automated deadline management and late submission detection
- Email and in-app notification system
- Responsive and user-friendly interface

This system enhances transparency, reduces administrative workload, and improves coordination between students and faculty.

2.2 Objectives of the System

- To develop a centralized digital platform for managing academic projects.
- To automate batch creation and mentor–student assignment.
- To implement customizable project stages with deadline tracking.
- To provide secure file upload and structured submission workflow.
- To enable faculty to review, approve, or reject submissions efficiently.
- To reduce manual documentation and administrative burden.
- To provide automated email and in-app notifications for important updates.
- To ensure secure role-based access control within the system.

2.3 User Requirements

1. Functional Requirements

Admin (Project Coordinator)

- Must be able to securely log in and manage profile information.
- Create and manage academic batches.
- Assign faculty mentors to batches.
- Bulk upload and manage student records using Excel.
- Create, modify, reorder, or delete project stages.
- Set and update submission deadlines.
- Monitor overall project progress and late submissions.
- View submission reports and system activity.

Faculty (Mentor)

- Must be able to securely log in and manage profile.
- View assigned batches and students.
- Download student submissions.
- Approve or reject documents.
- Provide remarks and suggestions.
- Track stage-wise student progress.

Student (Mentee)

- Must be able to securely log in and manage profile.
- View assigned mentor and batch information.
- View project stages and deadlines.
- Upload project documents in supported formats (PDF, DOC, PPT).
- Replace submissions before approval if required.
- View submission status and faculty remarks.
- Receive notifications regarding deadlines and approvals.

2. Non-Functional Requirements

Performance

- The system should load pages quickly and handle multiple users simultaneously.
- File uploads and downloads should execute efficiently.

Usability

- The interface should be simple, intuitive, and responsive.
- Users should easily navigate between dashboards and modules.

Security

- Only authenticated users should access the system.
- Role-based access control must prevent unauthorized data access.
- Uploaded files must be securely stored.

Maintainability

- The system should follow a modular architecture.
- Codebase should be scalable and easy to update.

Reliability

- The system should function consistently without data loss.
- Submission records must remain accurate and secure.

Compatibility

- The system should support desktop, tablet, and mobile devices.
- Compatible with modern web browsers such as Chrome, Edge, and Firefox.